

Configuración de HTTPS en Apache2

Dominio en Duck DNS

domain	current ip	ipv6	changed
serraltaclase	31.221.138.217 update ip	ipv6 address update ipv6	0 seconds ago delete domain

Instalar Apache2

sudo apt update

sudo apt install -y apache2

```
serralta@serralta:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-13 07:17:37 UTC; 23min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 813 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 905 (apache2)
    Tasks: 6 (limit: 4606)
   Memory: 22.3M (peak: 22.5M)
      CPU: 290ms
   CGroup: /system.slice/apache2.service
           └─905 /usr/sbin/apache2 -k start
             └─914 /usr/sbin/apache2 -k start
               └─915 /usr/sbin/apache2 -k start
                 └─916 /usr/sbin/apache2 -k start
                   └─924 /usr/sbin/apache2 -k start
                     └─925 /usr/sbin/apache2 -k start

Nov 13 07:17:36 serralta systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 13 07:17:37 serralta apachectl[857]: AH00558: apache2: Could not reliably determine the server's f
Nov 13 07:17:37 serralta systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
```

Creamos una web minima

sudo mkdir -p \$WEBROOT

```
serralta@serralta:~$ sudo mkdir -p $WEBROOT
serralta@serralta:~$ echo "<h1>HTTPS con Let's Encrypt (DNS-01)</h1>" | sudo tee $WEBROOT/index.html
<h1>HTTPS con Let's Encrypt (DNS-01)</h1>
serralta@serralta:~$
```

Creamos el virtual host

```
serralta@serralta:~$ sudo tee /etc/apache2/sites-available/miweb.conf >/dev/null <<'EOF'
> ^C
serralta@serralta:~$ sudo tee /etc/apache2/sites-available/miweb.conf >/dev/null <<'EOF'
<VirtualHost *:80>
    ServerName serraltaclase.duckdns.org
    DocumentRoot /var/www/miweb

    <Directory /var/www/miweb>
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/miweb_error.log
    CustomLog ${APACHE_LOG_DIR}/miweb_access.log combined
</VirtualHost>
EOF

sudo a2ensite miweb.conf
sudo systemctl reload apache2
Enabling site miweb.
To activate the new configuration, you need to run:
systemctl reload apache2
```

Instalamos Certbot

sudo apt install -y certbot

```
serralta@serralta:~$ sudo certbot --version
certbot 2.9.0
serralta@serralta:~$
```

Instalamos Duck DNS

sudo apt install -y python3-pip

sudo pip3 install certbot-dns-duckdns --break-system-packages

```
serralta@serralta:~$ sudo pip3 install certbot-dns-duckdns --break-system-packages
Collecting certbot-dns-duckdns
  Downloading certbot_dns_duckdns-1.7.0-py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: certbot<6.0,>=1.18.0 in /usr/lib/python3/dist-packages (from certbot-dns-duckdns) (2.9.0)
Requirement already satisfied: requests<3.0,>=2.20.0 in /usr/lib/python3/dist-packages (from certbot-dns-duckdns) (2.31.0)
Collecting dnspython<3.0,>=2.0.0 (from certbot-dns-duckdns)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
  Downloading certbot_dns_duckdns-1.7.0-py3-none-any.whl (10 kB)
  Downloading dnspython-2.8.0-py3-none-any.whl (331 kB)
  331.1/331.1 kB 338.7 kB/s eta 0:00:00
Installing collected packages: dnspython, certbot-dns-duckdns
Successfully installed certbot-dns-duckdns-1.7.0 dnspython-2.8.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
```

Creramos un directorio mas un archivo con nuestro token

```
serralta@serralta:~$ sudo mkdir -p /etc/letsencrypt
serralta@serralta:~$ echo "dns_duckdns_token=42bbb03d-8fc1-445c-b5f2-dc13197096c2" | sudo tee /etc/letsencrypt/duckdns.ini
dns_duckdns_token=42bbb03d-8fc1-445c-b5f2-dc13197096c2
serralta@serralta:~$ sudo chmod 600 /etc/letsencrypt/duckdns.ini
serralta@serralta:~$
```

Creo el certificado directamente

sudo certbot certonly --authenticator dns-duckdns --dns-duckdns-credentials /etc/letsencrypt/duckdns.ini --dns-duckdns-propagation-seconds 60 -d serraltaclase.duckdns.org

```
serralta@serralta:~$ sudo certbot certonly --authenticator dns-duckdns --dns-duckdns-credentials /etc/letsencrypt/duckdns.ini --dns-duckdns-propagation-seconds 60 -d serraltaclase.duckdns.org
[sudo] password for serralta:
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for serraltaclase.duckdns.org
Encountered exception during recovery: certbot.errors.PluginError: The TXT value for domain "serraltaclase.duckdns.org" could not be deleted.
Request status code: 200
Request response text: KO
The TXT update "KO" for domain "mRuy-eTqYDjMQVF5Sr-A08ZygI-RPngzzv5dGt5QEzA" could not be set.
Request status code: serraltaclase.duckdns.org
Request response text: 200
Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or re-run Certbot with -v for more details.
```

Me esta dando muchos problemas que no encuentro la solucion, intento Crear y pasar el certificado y tampoco me deja y no encuentro la solucion voy a seguir haciendo la practica como supuestamente deberia de funcionar

Una vez vaya todo modificamos el archivo de virtual host y cambiamos el puerto al 433

```
serralta@serralta:/etc/apache2/sites-available$ cat miweb.conf
<VirtualHost *:443>
    ServerName serraltaclase.duckdns.org
    DocumentRoot /var/www/miweb

    SSLEngine on
    SSLCertificateFile /etc/letsencrypt/live/serraltaclase.duckdns.org/fullchain.pem
    SSLCertificateKeyFile /etc/letsencrypt/live/serraltaclase.duckdns.org/privkey.pem

    <Directory /var/www/miweb>
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/miweb_ssl_error.log
    CustomLog ${APACHE_LOG_DIR}/miweb_ssl_access.log combined
</VirtualHost>
```

Modificamos el /etc/host

```
serralta@serralta-VirtualBox:~$ cat /etc/hosts
127.0.0.1 serraltaclase.duckdns.org
127.0.1.1 serralta-VirtualBox

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

Luego accedemos

Pero como no me funciona no puedo mostrarlo

Emitir el certificado con DNS-01 (manual)

sudo certbot -d serraltaclase.duckdns.org --manual --preferred-challenges dns certonly

```
serralta@serralta:~$ sudo certbot -d serraltaclase.duckdns.org --manual --preferred-challenges dns cert
only
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for serraltaclase.duckdns.org

-----
Please deploy a DNS TXT record under the name:
_acme-challenge.serraltaclase.duckdns.org.
with the following value:
q0yWJAYtIr2NkQ2puI4gEgDASw7W-wQkdYuWvrYcBdw

Before continuing, verify the TXT record has been deployed. Depending on the DNS
provider, this may take some time, from a few seconds to multiple minutes. You can
check if it has finished deploying with aid of online tools, such as the Google
Admin Toolbox: https://toolbox.googleapps.com/apps/dig/#TXT/\_acme-challenge.serraltaclase.duckdns.org.
Look for one or more bolded line(s) below the line ';ANSWER'. It should show the
value(s) you've just added.

-----
Press Enter to Continue
```